COL 703/7203

05th Nov.

Claim

Let H be a Herbrand structure and t be a ground term.

Then $H[\![t]\!] = t$

$H[\![c]\!] = c$ | by def^n

$H[\![f(t_1 \dots t_k)]\!] = f_H([\![t_1]\!], \dots, [\![t_k]\!])$

$= f_H(t_1, \dots, t_k)$ (by IH)

$= f(t_1 \dots t_k)$ (by def^n)

Claim: Let H be a Herbrand structure, ϕ be a formula, and t be a ground term

$$H \models \phi[t/x] \quad \text{iff} \quad H_{[x \mapsto t]} \models \phi$$

Translation Lemma

Theorem

Let $\phi = \forall x_1 \ldots \forall x_n \, \phi'$ be a closed formula in Skolem form.

Then ϕ is satisfiable iff it has a Herbrand model.

One direction: Trivial

(If ϕ has a Herbrand model, clearly ϕ is sat.)

Suppose $A \not\models \phi$

To prove that there is Herbrand model H such that $H \not\models \phi$.

To define H, all we need is the interpretation for all the predicates in ϕ.

$(t_1 .. t_k) \in P_H$ iff $A \not\models P(t_1 .. t_k)$

($t_1 .. t_k$: ground terms)

Proof: (by induction of the no. of quantifiers)

Base case: (No quantifiers) Since ϕ is closed, it must be a Boolean combination of atomic formulas $P(t_1 .. t_k)$ where $t_1 .. t_k$ are ground terms.

By construction A and H agree on all such atomic formulas.

$\therefore \quad H \models \phi$.

Inductive step.

Suppose $A \models \forall y\, \psi$

Claim: For all ground terms t, $A \models \psi[t/y]$

by IH: $H \models \psi[t/y]$ iff $H_{[y \mapsto t]} \models \psi$ for all ground terms t

iff $H \models \forall y\, \psi$

Why? Because: $A_{[y \mapsto A[\![t]\!]]} \models \psi$

Why? Translation lemma.

Ground Resolution Proof.

(a) (b) and (c) together entail (d)

Example

a) Everyone in the class is either sleepy or bored or day-dreaming.

b) All those who are bored are sleepy.

c) Someone in the class is not day-dreaming

d) Someone in the class is sleepy.

$$\forall x \left(C(x) \rightarrow S(x) \vee B(x) \vee D(x) \right)$$

$$\forall x \left(B(x) \rightarrow S(x) \right)$$

$$\exists x \left(C(x) \wedge \neg D(x) \right)$$

$$\neg \exists x \left(C(x) \wedge S(x) \right)$$

$\forall x \left(\neg C(x) \vee S(x) \vee B(x) \vee D(x) \right)$

$\forall x \left(\neg B(x) \vee S(x) \right)$

$C(a) \wedge \neg D(a)$

$\forall x \left(\neg C(x) \vee \neg S(x) \right)$

$\neg C(a) \vee S(a) \vee B(a) \vee D(a)$

$\neg B(a) \vee S(a)$

$C(a)$

$\neg D(a)$

$\neg C(a) \vee \neg S(a)$

$\neg C(a)$ $S(a)$ $D(a)$

$S(a)$ $D(a)$

$\neg C(a)$ $D(a)$

$D(a)$

□

Prove that this is valid

$$\exists x \forall y \; P(x,y) \Rightarrow \forall y \exists x \; P(x,y)$$

Exercise